

MATÍAS FERNÁNDEZ LAKATOS

PhD in Optical Physics & Msc in Theoretical Physics & MSc in Big Data

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Santiago de Compostela, España

ORCID



PROFESSIONAL EXPERIENCE

Research Engineer

- 2025 – present

Gradiant, Vigo, Spain
- Development of intelligent solutions for early threat detection in corporate cybersecurity.
 - Design and implementation of unsupervised machine learning models (β -Variational Autoencoder, Autoencoder, Extended and Isolation Forest) for real-time anomaly detection and behavior analysis in large-scale data environments.
 - Focus on model explainability (XAI) to enhance usability for non-technical users.
 - Integration into a scalable data architecture using technologies such as Kafka, Spark, and Airflow for efficient real-time data processing.

Researcher and Teaching Assistant at Udelar

- 2015 – 2024

Fac. Ingeniería, Udelar, Montevideo
- Taught university physics at the Physics Institute.
 - Conducted research in multiple projects, including study of long-range properties of the strong nuclear interaction, development of multispectral instruments for remote monitoring, phase object retrieval involving a new method and creating a new method of integration from just one derivative.

UNIVERSITY DEGREES

MSc Tecnologías de Análisis Masivo de Datos: Big Data

- 2024 – 2025

USC, Santiago de Compostela
- Thesis: Application of new anomaly detection models in Big Data contexts. Courses on Large-Scale Databases, Technologies for Managing Unstructured Information, Computing Technologies for Big Data, IoT, Statistics, Data Mining, Data Visualization, Business Intelligence, and Applications and Use Cases in Business.

PhD in Optics

- 2019 – 2024

Udelar, Montevideo
- Thesis: Visualization and Characterization of Phase Objects (clickable) Courses on Laboratory of fundamental electronics and scientific instrumentation, Coherent Optics, **Reinforcement Learning**, **Computer Image Processing**, **Computational Multivariate Statistics** and **Deep Learning for Computer Vision (cs231n)** . Advisors: PhD. José A. Ferrari, PhD Erna Frins and PhD Gastón A.Ayubi.

MSc in Quantum Chromodynamics

- 2016 – 2018

Udelar, Montevideo
- Thesis: Rol de los diversos acoplamientos en la cromodinámica cuántica infrarroja (clickable) Courses on Statistical Mechanics, Quantum Field Theory I and II, General Relativity. Advisors: PhD. Nicolás Wschebor and PhD Marcela Peláez.

Bachelor's Degree in Physics

- 2011 – 2015

Udelar, Montevideo

ABOUT ME

I am a physicist with postgraduate training (Ph.D. and MSc) and a Master's in Big Data Analytics. I currently work as a Research Engineer at Gradiant, developing intelligent systems for early anomaly and threat detection using machine learning and large-scale data processing. I have presented my research at conferences and scientific events, strengthening my technical communication and presentation skills. With over nine years of experience in teaching and research, I've cultivated a work style grounded in collaboration, empathy, and clear communication, integrating technical depth with a human-centered, multidisciplinary approach. Outside of work, I enjoy board games and reading, embracing the philosophy of "journey before destination" as a mindset for continuous learning and creative exploration.

ACHIEVEMENTS

Master's scholarship from the Comisión Académica de Posgrados (CAP, 2016)

PhD scholarship from the Agencia Nacional de Investigación e Innovación (ANII,2019)

PhD scholarship from the Comisión Académica de Posgrados (CAP,2022)

Member of the National System of Researchers in Uruguay (SNI, 2024).

4 first author peer-reviewed articles.

clickable, 2019, IJMPA Scopus
clickable, 2022, Optik Scopus
clickable, 2023, OLE Scopus
clickable, 2024, SPIE

Author peer-reviewed article

clickable, 2024, IOP Science
clickable, 2025, OLT

TECHNICAL SKILLS

Git

Data analysis

R

SQL

Python

Machine Learning

modeling

Matlab

Digital Image Processing

Linux

Statistical Analysis

LaTeX

Multivariate Analysis

Optimization

Database Management and Modeling